Chinese Journal of
POLYMER SCIENCE

RESEARCH ARTICLE

https://doi.org/10.1007/s10118-024-3235-0
Chinese J. Polym. Sci. **2024**, *42*, 1905–1914

Crosslinked Colorless Polyimide Films *via* Oxazole Groups as Crosslinking Agent: Preparation and Properties

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Electronic Supplementary Information

Abstract A series of transparent crosslinked colorless polyimide (CPI) films are prepared from 3,3′,4,4′-biphenyltetracarboxylic dianhydride (BPDA), 2,2′-bis(trifluoro-methyl)benzidine (TFMB), and 4,4′-oxydianiline (ODA) by thermal imidization, incorporating varying contents of 2,2'-(1,3-phenylene)bis(2-oxazoline) (1,3-PBO) as the crosslinking agent. Following the incorporation of the crosslinking structure, the CPI films show good optical transparency (approximately 85% winthin visible light range), enhanced glass transition temperature (from 325 °C to 341 °C), and improved thermal stability, and tensile strength. Notably, compared with the pristine uncrosslinked CPI, these crosslinked CPI films significantly increase in elongation at break (from 5.4% to 44.2%). Furthermore, the new approach ensures that crosslinked CPIs improve heat resistance and mechanical properties, while avoiding the embrittlement of materials. This study also offeres straightforward preparation methods for optically transparent crosslinked polyimides without additional processing steps. All these results make this approach can effectively improve the competitive performance of the CPI films for potential applications in microelectronic and optoelectronic fields.

Keywords Polyimide; Colorless transparent film; Crossking structure; Optical property; Thermal property; Mechanical property

Citation: Wang, H.; Liu, Z. C.; Shang, X.; Feng, L. D.; Bian, X. C.; Chen, X. S. Crosslinked colorless polyimide films *via* oxazole groups as crosslinking agent: preparation and properties. *Chinese J. Polym. Sci.* **2024**, *42*, 1905–1914.

INTRODUCTION

The rapid development of flexible electronics, including displays, solar cells, and wearables, has led to the increased use of polymer materials instead of inorganic glass.[1–8] This shift is attributed to the advantages of polymers, such as their lightweight nature, inherent flexibility, and cost-effectiveness. However, conventional polymers have limitations in meeting the demands of high-temperature processing (up to 400 °C) during device fabrication because of their relatively low glass transition temperature (T_g). Especially in the field of optoelectronics, polymers should maintain good optical transparency after undergoing high-temperature processes.[9–11] Therefore, significant research has been dedicated to the development of colorless transparent polymers with heat resistance.

Polyimides (PIs) are polymers with excellent heat resistance, mechanical properties, chemical stability, electrical insulation, and radiation resistance.[12,13] The above properties stem from the rigid and conjugated molecular structure of PIs, which could also cause strong inter- and intra-molecular charge transfer (CT) interactions between the electron-donating diamine units and the electron-accepting dianhydride units.[14,15] Therefore, this also results in conventional PIs appearing reddish-yellow or even dark brown, hindering their application as optical films in optoelectronics. Consequently, colorless polyimides (CPIs) have attracted considerable interest.

The discoloration of PIs can be mitigated by suppressing or diminishing the CT interactions between intra- and intermolecular. Efficient approaches include the introduction of alicyclic substituents,[16–19] highly electronegative fluorinated groups,[20–25] and asymmetrical or rigid noncoplanar substituents.[26,27] However, one of the most challenging aspects is the balance among their heat-resistant, optical, mechanical, and other properties for the molecular design of CPIs, as these objectives conflict with each other.[28]

To improve the comprehensive properties of CPIs, an effective strategy is to incorporate crosslinkable groups, such as acetenyl, norbornyl, acrylate, and benzocyclobutenyl, to create network structures.[29–35] In these studies, a post-polymerization process was necessary to create network structures. The final properties of CPIs depend on both the structure of the main chain and network structure.

In this work, our aim was to develop a method for synthesizing crosslinked CPIs with good transparency, heat resistance, and mechanical strength. The CPIs were synthesized with the primary monomers, 3,3′,4,4′-biphenyltetracarboxylic

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Received May 3, 2024; Accepted September 11, 2024; Published online November 11, 2024

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dianhydride (BPDA) and 2,2′-bis(trifluoromethyl)benzidine (TFMB), selected for their desirable properties, including reasonable T_g (>300 °C), low coefficient of thermal expansion (CTE), and good mechanical properties.[36–38] 2,2'-(1,3-Phenylene)bis(2- oxazoline) (1,3-PBO) was chosen as the crosslinking agent because of its high activity in the bifunctional group (two oxazoline rings) during the reaction.[39] We focused on investigating the influence of different crosslinking agent contents on CPI performance, including thermal, mechanical, and optical properties. We discuss the network structures of the crosslinked CPIs and the properties of the resulting films in detail.

EXPERIMENTAL

Materials

3,3′,4,4′-Biphenyltetracarbox-ylic dianhydride (BPDA, 99.5%), 4,4′-oxydianiline (ODA, 99.0%), 2,2′-bis(trifluoro-methyl)benzidine (TFMB, 99.9%), and 4-chlorophthalic anhydrid (99.9%) were purchased from Tianjin Chinatech Chemical CO., LTD. 2,2'-(1,3-Phenylene)bis(2-oxazoline) (1,3-PBO, 98.0%) was purchased from Shanghai Aladdin Bio-Chem Technology Co., Ltd. *N*-dimethylacetamide (DMAc) was purchased from Shanghai Macklin Biochemical Technology Co., Ltd. BPDA and ODA were purified by vacuum sublimation using an AnHui BEQ BOF-A-5-100 sublimation instrument. All the other reagents were used directly without further purification.

Preparation of CPI and Crosslinked CPI Films

CPI and crosslinked CPI films were prepared *via* a traditional two-step procedure, including poly(amide acid) (PAA) synthesis and CPI film preparation, as shown in Scheme 1.

PAA solutions were first synthesized. TFMB (19574.7 mg, 61.127 mmol), ODA (221.1 mg, 1.104 mmol), and DMAc (200 mL) were sequentially charged into a 500 mL three-necked flask equipped with a mechanical stirrer and nitrogen inlet/outlet. After dissolving the two types of diamines, an equimolar amount of BPDA (18492.7 mg, 62.853 mmol) was added to the flask. The reaction mixture was stirred for 48 h at room temperature to obtain a viscous, colorless, and transparent solution. The final concentration of solids was approximately 16 wt%. The resulting PAA solution was divided into five equal volumes. Different mass ratios of 1,3-PBO to CPI were added to the four PAA solutions separately and then stirred for 12 h at 23 °C.

CPI films were obtained using the following procedure: the PAA solutions were filtered off to remove insoluble impurities and then degassed under vacuum. They were doctor-bladed onto clean 0.7 mm glass plates. The plates were heated at 90 °C in an anaerobic oven for 10 h and then at 180, 240 and 330 °C for 1h in a nitrogen atmosphere. After natural cooling, the films were plunged into deionized water, peeled off from the plates, and dried at 120 °C for 4 h. Finally, uniform and flat films with a thickness of approximately 30 μm were obtained. CPI0 was derived from PAA solutions without a crosslinking agent. Four crosslinked CPI films were obtained, including CPI1–CPI4, which had different 1,3-PBO to CPI mass ratios of 0.1%, 0.5%, 1%, and 3% for CP1, CPI2, CPI3, and CPI4, respectively.

It should be noted that the introduction of ODA is intended to mitigate the variability in thin-film properties resulting from the different film preparation processes. CPI films based on BPDA/TFMB exhibit poor adhesion due to the presence of fluorinated groups, leading to spontaneous peeling during film preparation. Conversely, films containing ether bonds and 1,3-PBO crosslinking agents require manual peeling from the glass substrate to ensure experimental consistency.

To characterize the crosslinking structure more clearly, two model compounds were synthesized using 4-chlorophthalic anhydride and ODA: MC and CLMC. MC represents without 1,3-PBO and CLMC represents with a 1,3-PBO to CPI molar ratio of 1:2. The MC powders were prepared using the same procedure as the CPI films, as shown in Scheme S1 (in the electronic supplementary information, ESI).

Characterization

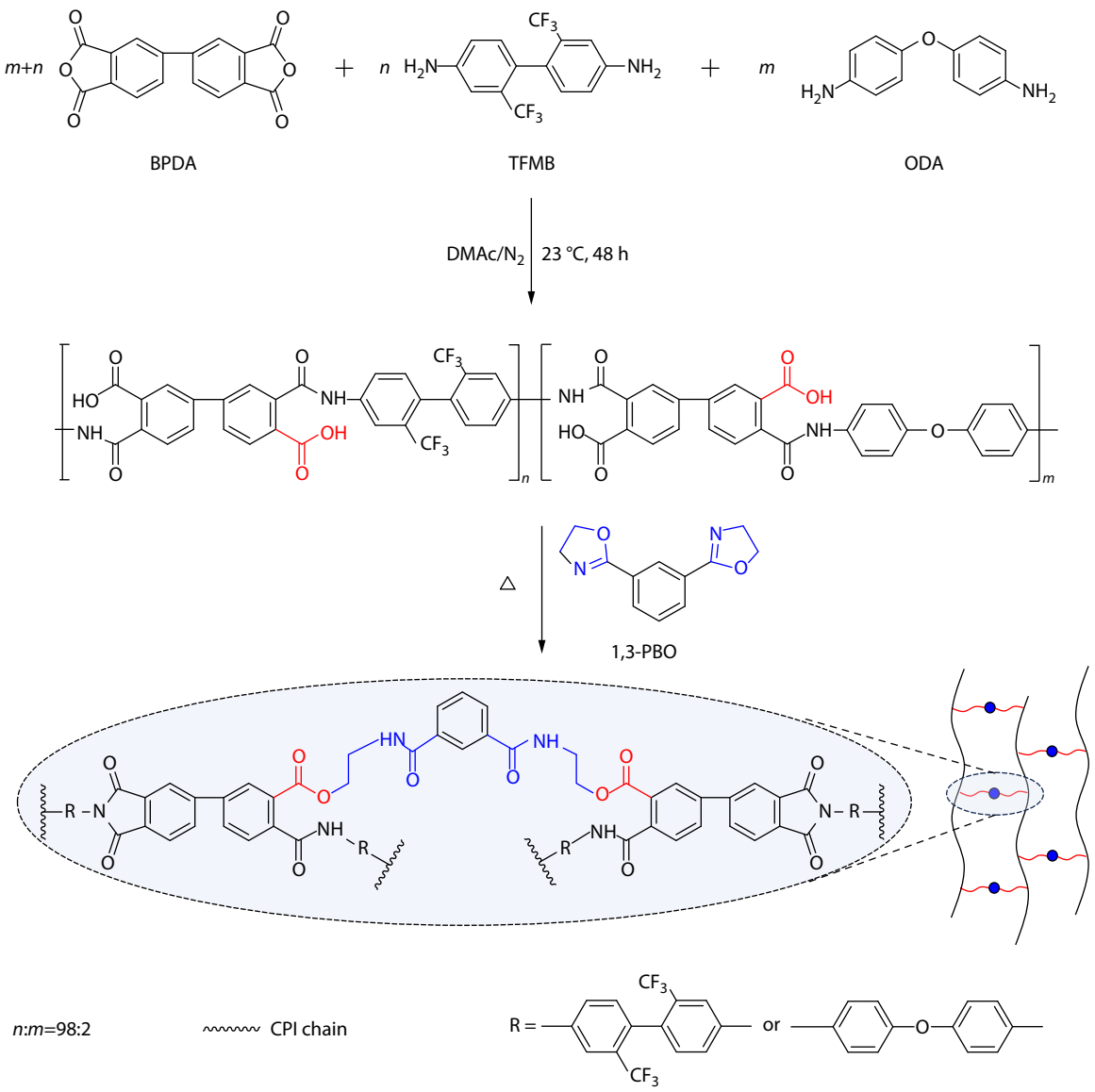
The dynamic viscosities of the PAA solutions were measured at 23 °C using a Shanghai Ni Run NDJ-5S digital rotary viscosimeter. The number-average molecular weight (M_n) and weight-average molecular weight (M_w) of the PAA polymers were determined by gel permeation chromatography (GPC) on an Agilent PL-GPC 50 instrument. Dimethylformamide (DMF) was used as the eluent at a flow rate of 1.0 mL/min. The densities of CPI films were measured using an Alfa Mirage SD-200L Electronic Densimeter.

Fourier-transform infrared (FTIR) spectra were recorded on a Bruker INVENIO-R spectrometer. Wide-angle X-ray diffraction (WAXD) was carried out on a Bruker D8 Advance X-ray diffractometer using Cu Kα radiation at a wavelength (λ) of 1.54 Å and a 2θ angle range from 5° to 60°. Solid-state nuclear magnetic state NMR (SSNMR) spectra were obtained on a Bruker AVANCE NEO 400 spectrometer, including nuclear magnetic resonance carbon spectroscopy (^{13}C-NMR). ^{1}H-NMR spectra of CLMC were obtained on a Bruker AVANCE spectrometer using dimethyl sulfoxide-d6 (DMSO-d_6) as the solvent.

Ultraviolet-visible (UV-Vis) spectra of the samples were recorded on a PerkinElmer Lambda 35 spectrophotometer in the transmittance mode in the wavelength range of 200–800 nm. The color intensities of the films were assessed using the Shenzhen Threenh NH310 Portable Colorimeter according to (International Commission on Illumination) standard D65 illuminant. The haze of CPI films was measured using a Shanghai Optical WGW Haze Meter.

Solvent resistance tests were performed to verify the formation of the crosslinking structures. CPI films (approximately 35 mg) were immersed in DMAc (6 g) for 48 h at 23 °C. The residual films were then dried at 160 °C in a vacuum oven for 12 h to remove the solvent. The solvent resistance of the films was characterized using the residual mass ratio, which is the ratio of the residual mass to the original mass of the CPI films. The water absorption was tested by weighing the weight changes before and after immersion in water at 23 °C for 24 h.

Thermogravimetric analysis (TGA) of the CPI films was performed on a TA Q500 thermal analysis system at a heating rate of 10 °C/min in a nitrogen atmosphere from 90 °C to 800 °C. Dynamic mechanical analysis (DMA) was performed using a TA DMA850 instrument with a heating rate of 3 °C/min and a frequency of 1 Hz from 70 °C to 400 °C. Thermomechanical analysis (TMA) measurements were carried out on a TA Q400



Scheme 1 Synthesis of crosslinked CPIs.

thermal analysis system with a fixed load of 0.05 N at a heating rate of 5 °C/min under N_2 flow. The coefficient of thermal expansion (CTE) of the films was estimated in the temperature range of 50–200 °C. Prior to starting the measurement, the samples were heated to 230 °C to eliminate residual stress.

Tensile testing was performed using an MTS E44 electromechanical test system at a constant displacement rate of 10 mm/min. Five samples were tested for each CPI film and the average values were recorded.

RESULTS AND DISCUSSION

Preparation and Characterization of Films

As shown in Scheme 1, a series of CPI films was prepared *via* a two-step procedure with PAA as the precursor, followed by thermal imidization at an elevated temperature of 330 °C. Crosslinked CPI films can be obtained by reacting carboxyl groups in PAA with oxazole groups in the crosslinking agent to form networks during the subsequent PAA imidization for the preparation of CPIs. The dynamic viscosities (μ) and molecular weights of the PAAs are summarized in Table 1. The M_n and μ values of these PAAs were approximately 300 kg/mol and 9000–9500 mPa·s, respectively. These results indicate that all PAAs possess high molecular weights and moderate viscosities, which makes it possible to obtain tough, free-standing, and transparent films cast from their DMAc solutions. The results, with little difference, also indicate that there is no crosslinking reaction in the PAAs with the crosslinking agent at room temperature, which are stable.

The FTIR spectra of CPI films and 1,3-PBO powders are shown in Fig. 1(a). For the CPI films, the characteristic bands of the asymmetric and symmetric stretching vibrations of the carbonyl on the imide ring are at approximately 1778 and 1720 cm^{-1}. Characteristic bands of C—N—C in the imide ring can be observed at approximately 1360 cm^{-1}. At the same

Table 1 Dynamic viscosities (μ) and molecular weights of PAAs.

PAA	μ [a] (mPa·s)	M_n [b] (kg/mol)	M_w [b] (kg/mol)	PDI [b]
CPI0	9020	302	402	1.33
CPI1	9470	297	410	1.38
CPI2	9170	305	400	1.31
CPI3	9310	299	407	1.36
CPI4	9500	299	409	1.37

[a] Measured at a concentration of 16 wt% DMAc at 23 °C; [b] Measured by GPC.

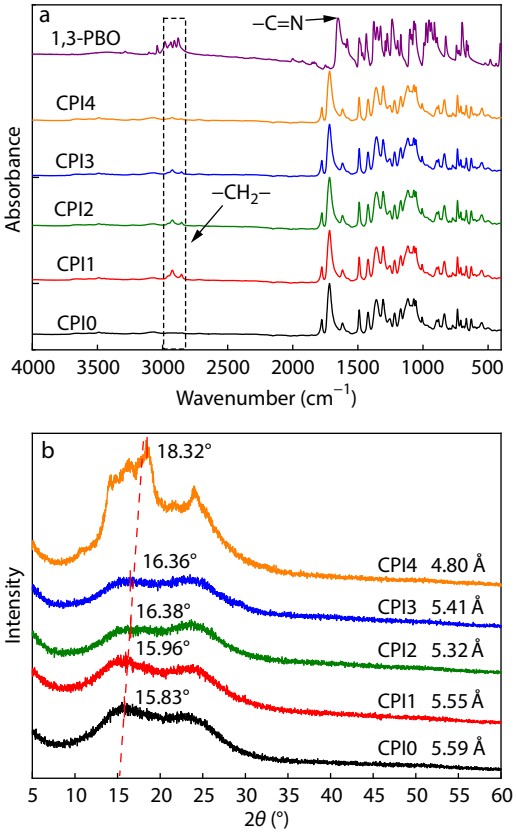


Fig. 1 (a) ATR-FTIR spectra and (b) WAXD patterns of CPI films.

time, there is no obvious N—H adsorption band in the range of 1600 cm^{-1} to 1700 cm^{-1}, indicating that the obtained CPIs are completely imidized. Strong C—F characteristic absorption bands can be observed at approximately 1307 cm^{-1}. For 1,3-PBO, the characteristic bands of C=N in the oxazole ring can be observed at approximately 1652 cm^{-1}. Moreover, the characteristic bands of C=N disappear, and the obvious characteristic bands of —CH_2— vibration appear near 2924 and 2854 cm^{-1} for the crosslinked CPIs (CPI1–CPI4), which are caused by the ring-opening reaction between the oxazole groups in the crosslinking agents and the carboxyl groups on the molecular chain of the PAAs.[40] The disappearance of C=N and the presence of —CH_2— indicate that the crosslinking agent effectively causes crosslinking between CPI molecular chains to form a network structure.

The FTIR spectra of MC, CLMC, and 1,3-PBO powders are shown in Fig. S1 (in ESI). For CLMC, the obvious characteristic bands of —CH_2— vibration and —NH appear near 2936 and 3482 cm^{-1}, the CLMC, and the characteristic bands of C=N in the oxazole ring disappeared at around 1652 cm^{-1}. In contrast, no characteristic bands of —CH_2— and —NH were observed in MC.

The ^{13}C-NMR spectra of MC, CLMC, and 1,3-PBO powders are presented in Fig. S2 (in ESI). All aromatic carbon signals appear in the range of 112–156 ppm. For MC and CLMC, carbonyl carbon signals were observed at approximately 166 ppm (Figs. S2a and S2b in ESI). The carbonyl carbon in the ester group generated by the crosslinking reaction could not be clearly distinguished from the carbonyl carbon in the imide ring, mainly because of the similar chemical shift effects exerted by the tertiary amine nitrogen and —O— in the ester group on the carbonyl carbon. For 1,3-PBO, the C=N carbon signals in the oxazole ring appeared at approximately 163 ppm (Fig. S2c in ESI), but are absent in MC and CLMC. The methylene carbon signals can be observed at 37–48 ppm in CLMC and at 55–69 ppm in 1,3-PBO, which cannot be found in MC. It is worth noting that the chemical shift of the methylene carbon in CLMC shows significant changes compared to 1,3-PBO. This is mainly due to significant changes in the functional groups connected to the methylene group, resulting from the ring-opening reaction between the oxazole and carboxyl groups, leading to the formation of ester and imine groups. The FTIR and ^{13}C-NMR spectra confirmed the designed chemical structures in the crosslinking samples. Fig. S3 (in ESI) presents the solution ^{1}H-NMR spectrum of CLMC. The proton peaks in the 8.1–7.0 ppm range are assigned to aromatic protons. The characteristic peak of the —CH_2— protons is observed around 5.3 ppm, while the peak at around 10.0 ppm is associated with the —NH protons. Notably, the absence of proton peaks at approximately 4 ppm, typically assigned to —OH groups, suggests that hydroxyl groups are absent in the final structure. These observations indicate that the ester groups generated during the crosslinking reaction were not decomposed and transformed into imide rings during the thermal imidization process. Furthermore, the introduction of crosslinking agents effectively results in the formation of stable crosslinked structures.

To investigate the molecular packing and better understand the relationship between the CPI structures and properties, the morphology of the CPI films was analyzed using WAXD. As clearly seen from the WAXD patterns depicted in Fig. 1(b), all CPI films show a broad diffraction peak from 14° to 28°, indicating that these CPIs are amorphous. The peak at 14°–19° originates from the average interchain distances, which are calculated according to Bragg's equation: $n\lambda=2d\sin\theta$. The average interchain instances (d-spacing) of the CPIs, as summarized in Table 2, ranged from 4.80 Å to 5.59 Å, decreasing with the increase of 1,3-PBO contents. This result suggests that the network structure results in a dense stacking density of the CPI molecular chains. Moreover, the diffraction peak near 24° represents the π-π stacking interaction distance between the benzene rings, approximately 3.80 Å, which hardly changes with the monomer composition. However, the intensity of this peak increased with the degree of crosslinking, indicating an enhanced π-π stacking interaction. Furthermore, the distinct sharp diffraction peaks in the CPI4 curve indicate partial crystallization within the

Table 2 Density and average interchain instances (*d*-spacing) of CPI films.

CPI	Density (g/cm^3)	*d*-Spacing (Å)
CPI0	1.30	5.59
CPI1	1.41	5.55
CPI2	1.53	5.32
CPI3	1.43	5.41
CPI4	1.62	4.80

polymer.[36] This occurs because the reaction between 1,3-PBO and the carboxyl groups in the PAA main chain not only forms crosslink points but also partially destroys the rigid imide rings in the CPI molecular chains, thereby increasing the flexibility of the molecular chains. When the content of the crosslinking agent is high, more imide rings are destroyed, shortening the rigid segments and making them easier to arrange in an orderly and structured manner, thus resulting in partial crystallization phenomena.[41]

The densities of the CPI films, as summarized in Table 2, range from 1.30 g/cm^3 to 1.62 g/cm^3, increasing with the increase of 1,3-PBO contents. The increase in the film density corresponds to a decrease in the molecular free volume fraction, resulting in denser chain stacking. The density test results for the CPI films were consistent with the WAXD test results.

Solvent Resistance Water Absorption

To confirm the existence of the crosslinking structures, DMAc dissolution experiments were performed. Fig. 2(a) shows the residual mass ratio over the 1,3-PBO content. The CPI0 film, absent of crosslinking agent, completely dissolved in DMAc after immersion in DMAc for 48 h. Conversely, the CPI1-CPI4 films exhibited progressively higher proportions of undissolved material with increasing 1,3-PBO content. Notably, the CPI3 film displayed remarkable resistance to DMAc, retaining 99.7% of its original weight even after 48 h of immersion. This observation suggests a high degree of cross-linking within the films. Although the CPI4 film maintained a high residual mass ratio of 95.4%, it displayed a slight decrease compared to CPI3. This phenomenon can be attributed to the presence of single bonds formed during the ring-opening reaction of the oxazole groups in excess 1,3-PBO. These findings collectively confirm the successful introduction of crosslinking structures by 1,3-PBO. Furthermore, the results suggest an initial increase in the crosslinking degree of CPI films with increasing 1,3-PBO content, followed by a slight decrease at higher crosslinking agent content.

The water absorption rates of the CPI films were also measured by weighing the films before (W_0) and after (W_1) water absorption and were calculated using Eq. (1):

$$\text{Water absorption rate (\%)} = (W_1 - W_0)/W_0 \times 100\% \quad (1)$$

Fig. 2(b) illustrates the water absorption rates, demonstrating good hydrophobicity, with rates ranging from 0.58% to 0.70%. Crosslinked CPI films exhibit higher water absorption rates than uncrosslinked films, which is attributed to the formation of amide bonds through the ring-opening reaction of oxazole groups. Despite the introduction of amide bonds, which are known for their water absorption capability,[42,43] all CPI films maintained water absorption rates below 0.9% owing to the high fluorine content.

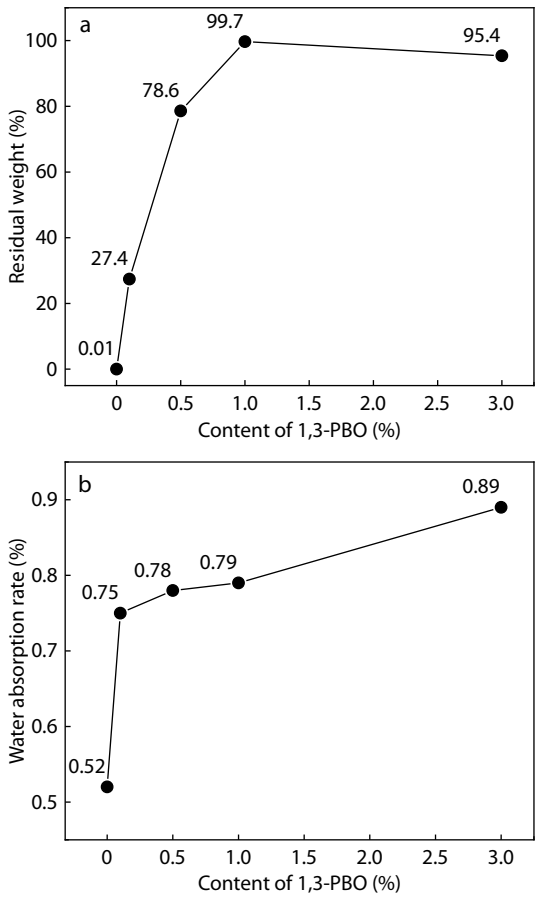


Fig. 2 (a) Residual mass ratio of CPI films before and after crosslinking in DMAc; (b) Water absorption rate of CPI films with different amounts of 1,3-PBO.

Optical Properties

Fig. 3(a) shows the visual appearances of the CPI films on white printed paper with a blue logo. All CPI films exhibited high transparency with a light yellowish tint, except for CPI4, which showed slight opacity. The UV-Vis-NIR spectra are shown in Fig. 3(b), and the results are summarized in Table 3, which provides further insight into the optical properties. The transmittance at 450 nm (T_{450}) for the CPI0 to CPI3 films was greater than 81%, and the average transmittance (T_{Vis}) within the visible light range (400–700 nm) was greater than 84%. As the crosslinking agent content increased, the transmittance (T_{450} and T_{Vis}) values of these films decreased. This trend was particularly pronounced in CPI4, where the transmittance (only T_{450} was 65% and T_{Vis} was 77%) significantly decreased.

The color intensity of the films was assessed using a colorimeter. Some parameters include lightness (L^*), yellowness (b^*), redness (a^*), and yellow index (YI). As shown in Fig. 3(c) and Table 2, the CPI films showed high L^* values of over 95, indicating good lightness. They also showed a^* values of approximately −1.0, b^* values of 4.2–14.7, and yellowness indices (YI) of 7.0–25.4. With an increase in the crosslinking agent content, CPI films exhibit reduced L^* but increased b^* values, especially CPI4, indicating deepening of the film color. Additionally, CPI0-CPI3 possesses a low haze of 0.8–1.4, demonstrating a good degree of clarity. CPI4 has a high haze

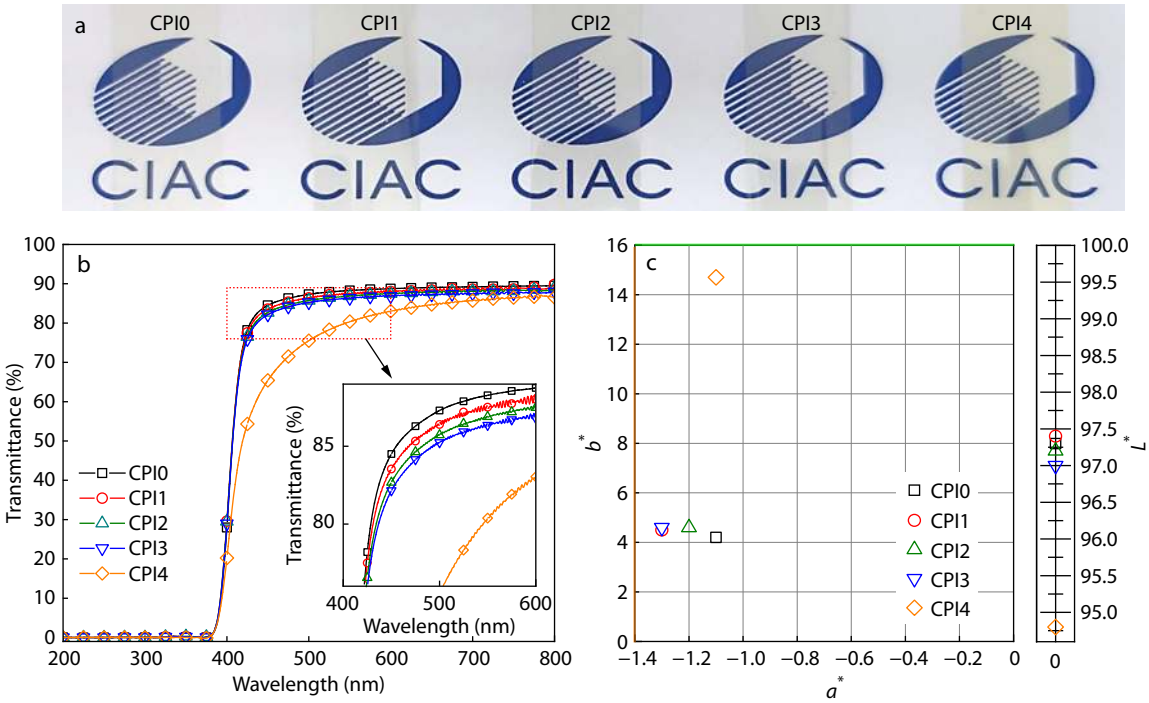


Fig. 3 (a) Visual appearance, (b) UV-Vis-NIR spectra and (c) 2D image of CIE Lab color parameters of various CPI films.

Table 3 Optical properties of CPIs.

CPI	T_{450} [a] (%)	T_{vis} [b] (%)	L^* [c]	a^* [c]	b^* [c]	YI [c]	Haze (%)
CPI0	84.5	85.9	97.3	−1.1	4.2	7.0	0.8
CPI1	82.6	85.3	97.4	−1.3	4.5	7.4	0.9
CPI2	82.5	84.7	97.2	−1.2	4.6	8.1	1.0
CPI3	81.8	84.1	97.0	−1.3	4.6	9.4	1.4
CPI4	65.2	77.0	94.8	−1.1	14.7	25.4	8.0

[a] Transmittance at 450 nm; [b] Transmittance in the visible light range of 400-700 nm. [c] L^*, a^*, b^*, YI: CIE Lab optical parameters.

of 8.0 due to increased light scattering inside the film caused by partial crystallization, resulting in poor clarity. For CPI0 to CPI3, the optical properties showed minimal variation with increasing crosslinking-agent content. However, CPI4 exhibited a significant decline in optical performance. The underlying reasons for these observations can be elucidated using the WAXD analysis. As the content of the crosslinking agent increased, closer interchain distances and enhanced π-π stacking interactions occurred, resulting in film yellowing (indicated by b^* and YI) and reduced transmittance.[15,24] At lower crosslinking agent contents (for CPI0 to CPI3), these effects on the optical performance are not significant. However, at higher concentrations (such as in CPI4), these effects become more pronounced because interchain distance sharply decreases (from 5.41 Å to 4.80 Å) and π-π stacking interaction significantly increases. Moreover, partial crystallization is observed because the shortening of rigid chain segments and an increase in flexible chain segments may cause rigid chain segments to fold and become more orderly arranged, which further exacerbates the deterioration of optical performance.

Thermal Properties

The thermal properties of the CPIs were characterized using TGA, DMA, and TMA, as illustrated in Fig. 4 and summarized in Table 4. As shown in Fig. 4(a), the temperature of 5% weight loss ($T_{D5\%}$) of the CPIs was determined by TGA in N_2 with the the corresponding results are summarized in Table 4. With the exception of CPI4, CPI films exhibited favorable thermal stability, with $T_{D5\%}$ values ranging from 536 °C to 551 °C. $T_{D5\%}$ show a trend of initially increased and then decreased with increasing crosslinking agent content. CPI4 experiences approximately 7% weight loss within 350–380 °C, which is attributed to the decomposition of single-bond groups generated during the crosslinking reaction. The thermo-mechanical responses of the CPIs were evaluated by DMA, with the temperature dependence of tanδ depicted in Fig. 4(b). T_g was determined from the peak temperatures of the tanδ curves, and the detailed results are listed in Table 4. T_g exhibited a slight increase with increasing crosslinking agent content, ranging from 325 °C to 341 °C for CPI0-CPI2. However, T_g decreased with a further increase in the content of the crosslinking agent. The variations observed in the thermal decomposition temperature and T_g were attributed to the combined effect of the formation of crosslinked network structures and the degree of disruption of the rigid imide ring structures in the main chain resulting from the crosslinking reaction. At lower levels of the crosslinking agent, the disruption of imide rings is limited; thus, the enhanced heat resistance is primarily governed by the formation of the crosslinked network structure. However, as the content of the crosslinking agent further in-

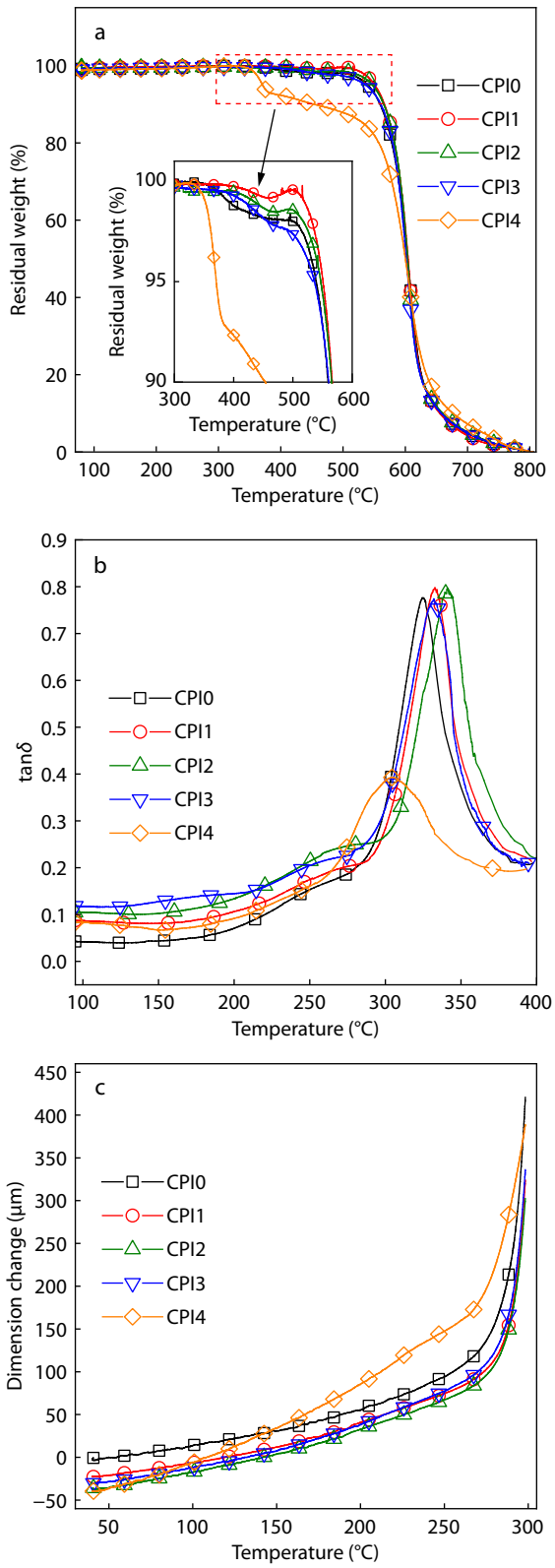


Fig. 4 (a) TGA, (b) DMA and (c) TMA thermograms of CPI films.

creased, the degree of disruption of the rigid structures within the main chain intensified, resulting in the generation of an increased number of single-bond groups. Consequently, the predominance of the disruption of rigid structures significantly impacts the heat resistance, ultimately leading to a decrease in the overall heat resistance. Meanwhile, apparent sub-T_g relaxation (β relaxation) is also observed at 250–270 °C for CPIs in Fig. 4(b), which is attributed to the local motion of the single bonds generated by the ring-opening reaction of the oxazole groups in the crosslinking agent. The β relaxation in the CPIs increased as the crosslinking agent content increased. This can be ascribed to the introduction of crosslinking agents, which disrupt the rigid structure of the CPI main chain, making chain segments more flexible. The dimensional stabilities of all CPI films were characterized by TMA (Fig. 4c), and the detailed results are shown in Table 4. The CTE values range from 16.7 ppm/°C to 22.4 ppm/°C between 50 and 200 °C. The increase in the CTE is attributed to the disruption of more imide rings as the crosslinking agent content increases. The oxazole groups of the crosslinking agents react with carboxyl groups on the CPI backbone, leading to ring-opening reactions that not only partially disrupt the rigid imide ring structure, but also introduce flexible groups such as esters, amines, and methylene groups. This process significantly enhances the flexibility of the CPI chains. Therefore, the CTE of the crosslinked CPIs exhibits an upward trend with increasing crosslinker content, which is consistent with the results of the presence of $-CH_2-$ in FTIR and β relaxation in DMA.

Table 4 Thermal properties of CPI films.

CPI	$T_{D5\%}$ [a] (°C)	T_g [b] (°C)	CTE [c] (ppm/°C)
CPI0	539	325	16.7
CPI1	547	333	18.8
CPI2	551	341	20.8
CPI3	536	332	20.5
CPI4	371	305	24.4

[a] Temperature at 5% weight loss, measured by TGA in nitrogen at a heating rate of 10 °C/min; [b] Glass transition temperature measured by DMA at a heating rate of 3 °C/min at 1 Hz; [c] Coefficient of thermal expansion measured in the range of 50–200 °C at a heating rate of 5 °C/min.

Mechanical Properties

The tensile testing curves of CPI films with different 1,3-PBO contents are shown in Fig. 5. The tensile strength (σ_m), tensile modulus (E_t) and elongation at break (ε_b) values of the films are listed in Table 5. As the content of the crosslinking agent increased, the tensile strength of the films first increases from 143.2 MPa to 197.6 MPa, then decreases to 170.8 MPa, while the tensile modulus decreases from 3.8 GPa to 2.9 GPa, and then increases to 3.4 GPa. The variation in the tensile strength is attributed to the same factors as the changes in T_g, resulting from the combined effect of the formation of crosslinked network structures and the degree of disruption of the rigid imide ring structures in the main chain. The change in the tensile modulus is the result of the disruption of the rigid structures in the main chain, and the increase in the modulus observed in CPI4 is attributed to the occurrence of partial crystallization. It is worth noting that with the increase in crosslinking agent content, there was a significant increase in the elongation at break of the films, from 5.4% to 44.2%. This can be attributed to the conversion of the rigid imide rings into flexible groups through crosslinking reactions, rendering the CPI molecular chains more flexible.

https://doi.org/10.1007/s10118-024-3235-0

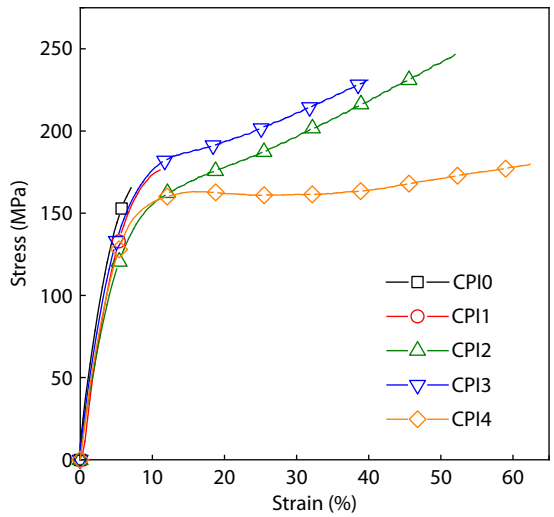


Fig. 5 Stress-strain curves of CPI films.

Table 5 Mechanical properties of CPI films.

CPI	σ_m [a] (MPa)	E_t [a] (GPa)	ε_b [a] (%)
CPI0	143.2	3.8	5.4
CPI1	172.1	3.6	10.6
CPI2	197.6	3.1	28.6
CPI3	187.8	2.9	27.9
CPI4	170.8	3.4	44.2

[a] Five samples were tested for each CPI, and the average values were used for data analysis.

Comprehensive Performance

Considering the influence of the crosslinking agent 1,3-PBO on the optical, thermal, and mechanical properties of the BPDA/TFMB/ODA copolymers, their comprehensive performance was characterized using a radar chart, as shown in Fig. 6. The results indicated the following trend in the comprehensive performance of the copolymers: CPI2 > CPI3 > CPI1 > CPI0 > CPI4. Notably, CPI2 exhibited the best comprehensive performance, indicating that the incorporation of an optimal amount of 1,3-PBO can achieve high-performance CPI films. The reasons for this are as follows:

The excellent thermal and mechanical properties of PI are mainly due to the presence of highly stable and rigid imide rings in the molecular chains as well as strong inter-molecular interactions. Furthermore, the formation of a crosslinked network structure can reduce the mobility of molecular chains. However, unlike previous research, where crosslinking agents react with side chain groups to form a crosslinked network structure, the current approach utilizes a crosslinking agent to react with the carboxyl groups on the PAA main chain, resulting in the generation of flexible moieties, such as ester, methylene, and amide groups (as shown in Scheme 1). Although this approach effectively generates a crosslinked network, it can also partially disrupt the rigidity of the main chain. This tradeoff between promoting crosslinking and maintaining chain rigidity is critical for optimizing the comprehensive performance of CPI films. In CPI2, the 1,3-PBO content appears to strike a balance, enabling the formation of a crosslinked network while preserving an adequate degree of chain stiffness. This synergy between crosslinking and flexibility is believed to contribute to the superior comprehensive performance observed in the CPI2 films. Additionally, we conducted comparisons with certain commercial products. For instance, when evaluating CPI2 against Mitsubishi Gas Chemical's Neopulim™ S200,[44] CPI2 exhibited superior properties in most aspects, except for optical properties (yellow index: 8.1 versus 1.3; visible light transmittance: 85% versus 90%). Conversely, when compared with Kolon's CPI™ Grade B,[45] the thermal performance was comparable, but significant differences were observed in other properties. These differences can be attributed to the optical performance being primarily influenced by the inherent optical characteristics of the CPI based on BPDA/TFMB/ODA with a YI of 7.0. Therefore, this approach not only improves the comprehensive performance of the crosslinking structure, but also increases the flexibility of the material.

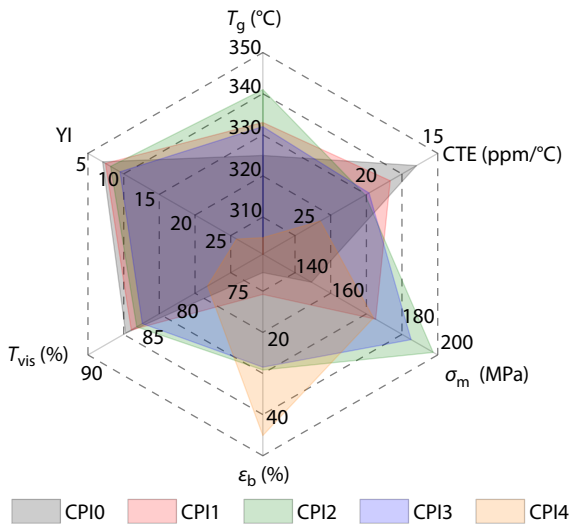


Fig. 6 Evaluation of comprehensive performance for CPI films based on the results listed in Tables 3–5.

CONCLUSIONS

In summary, a series of crosslinked CPIs was successfully synthesized by the incorporation of a 1,3-PBO crosslinking agent into the BPDA/TFMB/ODA PAA solution. The introduction of 1,3-PBO facilitated the formation of a crosslinked structure during thermal imidization. When an appropriate amount of the crosslinking agent was added, the optical performance of the film did not significantly decrease, as evidenced by the T_{vis} value of approximately 85%. Significant improvements in the thermal and mechanical properties were observed. T_g increases from 325 °C to 341 °C, and the tensile strength improves from 143.2 MPa to 197.6 MPa. Additionally, the method used in this study can adjust the dimensional stability and significantly enhance the elongation at break, increasing from 5.4% to 44.2%. The observed trends were analyzed and the underlying mechanisms were elucidated. The crosslinking process involves a reaction between the oxazole groups in the crosslinking agent and the carboxyl groups within the PAA molecular chains. This reaction leads to partial disruption of the rigid imide rings in the final PI main chain, accompanied by the generation of flexible groups. The comprehensive performance of the resultant CPI films

stems from the interplay between the formation of crosslinked network structures and the extent of disruption of the rigid imide rings in the main chain. This study provides a simple method for preparing crosslinked CPIs without additional processing steps. The prepared crosslinked CPI films demonstrated the effectiveness of this method in enhancing the comprehensive performance of CPI films. However, it is important to note that the CPI films in this study showed significant gaps in performance for display substrates, which is a major limitation for the widespread application of CPI in optoelectronics. Nonetheless, the method proposed in this study effectively enhanced the thermal and mechanical properties of CPI. With the development of improved CPI formulations in the future, this method holds significant potential for applications in microelectronics and optoelectronics.

Conflict of Interests

Xue-Si Chen is an editorial board member for *Chinese Journal of Polymer Science* and was not involved in the editorial review or the decision to publish this article. All authors declare that there are no competing interests.

Electronic Supplementary Information

Electronic supplementary information (ESI) is available free of charge in the online version of this article at http://doi.org/10.1007/s10118-024-3235-0.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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